MATH 231 – Applied Linear Algebra

Lecture 0 – Foundations Review

A self-study primer to read *before* the course begins

Fall 2026

Before you start

This lecture is **Lecture 0**: it contains no new linear algebra. Everything here is material you have seen before, gathered in one place so that on Day 1 we can speak a common language. Your only formal prerequisite is **MATH 141** (Calculus I). If a page feels like review, skim it; if a page feels new, slow down and work the examples with a pencil — that is the whole point of a Lecture 0.

How to use this document

Read actively. For every *Example*, cover the solution and try it first. The **Looking ahead** notes tell you *why* a piece of review matters — each points to the unit of MATH 231 where the idea returns with full force. A short self-check with answers is at the end.

Contents

1. Numbers and Sets

1.1 The number systems

We build up from the counting numbers to the real line, each system fixing a shortcoming of the previous one.

$$\underbrace{\mathbb{N}}_{\text{naturals}} \subset \underbrace{\mathbb{Z}}_{\text{integers}} \subset \underbrace{\mathbb{Q}}_{\text{rationals}} \subset \underbrace{\mathbb{R}}_{\text{reals}} \subset \underbrace{\mathbb{C}}_{\text{complex}}$$

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ (in this course we include 0; some books start at 1).
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ — adds subtraction.
- $\mathbb{Q} = \left\{\frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0\right\}$ — adds division.
- $\mathbb{R}$ — fills the "gaps" (e.g. $\sqrt{2}$, π, e are real but not rational).
- $\mathbb{C} = \{a + bi : a, b \in \mathbb{R},\ i^2 = -1\}$ — adds roots of *every* polynomial.

Looking ahead. *A vector space (Unit 1) is always built* over *a number system called its field of scalars — almost always $\mathbb{R}$ or $\mathbb{C}$. Complex numbers get their own treatment in Unit 3, where $i^2 = -1$ becomes rotation by $90°$.*

1.2 Sets, elements, and membership

A **set** is an unordered collection of distinct objects called its **elements**. We write:

$$x \in A \quad (\text{"}x \text{ is an element of } A\text{"}), \qquad x \notin A \quad (\text{"}x \text{ is not an element of } A\text{"}).$$

Two ways to describe a set:

- **Roster (list) form:** $A = \{2, 4, 6, 8\}$.
- **Set-builder form:** $A = \{\, x : \text{condition on } x \,\}$, read "the set of all x *such that* …". The colon ":" (or "|") means *such that*.

Example 1.1. Write "the even numbers between 1 and 9" in set-builder form.

$$A = \{\, x \in \mathbb{Z} : 1 \leq x \leq 9 \text{ and } x \text{ is even} \,\} = \{2, 4, 6, 8\}.$$

The part before the colon says *what kind* of object ($x \in \mathbb{Z}$); the part after says *which ones* we keep.

Example 1.2. The unit circle is a set of points: $S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$. This is the same object as the geometric circle of §7 — a *set* and an *equation* describing the same thing.

The **empty set** $\varnothing = \{\ \}$ has no elements. The number of elements of a finite set A is its **cardinality** $|A|$; e.g. $|\{2, 4, 6, 8\}| = 4$.

Looking ahead. *Set-builder notation is how we will* define *the central objects of the course: a span, a subspace, a null space, and the feasible region of an optimization problem are all written* $\{\, x : \textit{some linear condition} \,\}$.

1.3 Subsets and equality

A is a **subset** of B, written $A \subseteq B$, if every element of A is also in B. If in addition $A \neq B$ we call it a **proper subset**, $A \subsetneq B$. Two sets are **equal** exactly when each contains the other:

$$A = B \iff (A \subseteq B \text{ and } B \subseteq A).$$

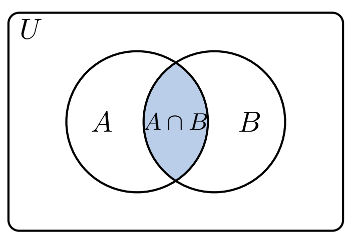
This "two-way inclusion" is the standard way to *prove* two sets are equal.

Example 1.3. Let $A = \{1, 2\}$ and $B = \{1, 2, 3\}$. Then $A \subseteq B$ (indeed $A \subsetneq B$), but $B \not\subseteq A$ since $3 \in B$ but $3 \notin A$. Also $\varnothing \subseteq A$ for *every* set A.

1.4 Set operations

Given sets A, B inside some universe U:

Union	$A \cup B = \{x : x \in A \text{ or } x \in B\}$	everything in either
Intersection	$A \cap B = \{x : x \in A \text{ and } x \in B\}$	the overlap
Difference	$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$	in A but not B
Complement	$A^c = U \setminus A = \{x \in U : x \notin A\}$	everything outside A



Example 1.4. Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, universe $U = \{1, 2, \ldots, 8\}$.

$$A \cup B = \{1, 2, 3, 4, 5, 6\}, \quad A \cap B = \{3, 4\}, \quad A \setminus B = \{1, 2\}, \quad A^c = \{5, 6, 7, 8\}.$$

The **Cartesian product** pairs elements from two sets in order:

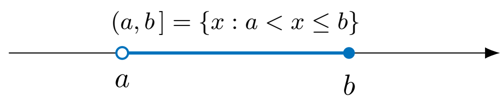
$$A \times B = \{(a, b) : a \in A,\ b \in B\}.$$

The plane is $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x, y \in \mathbb{R}\}$ and space is $\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$. Order matters: $(1, 2) \neq (2, 1)$.

Looking ahead. $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$ *(n copies) is where* vectors *live (Unit 1). A vector is nothing more exotic than an ordered list of n real numbers — an element of a Cartesian product.*

1.5 Intervals and the real line

Intervals are the most common subsets of $\mathbb{R}$. They come in open, closed, and half-open flavors; a filled dot ($\bullet$) means the endpoint is included, an open dot ($\circ$) means it is excluded.



$[a, b] = \{x : a \leq x \leq b\}$		closed (both ends in)
$(a, b) = \{x : a < x < b\}$		open (both ends out)
$[a, b) = \{x : a \leq x < b\}$,	$(a, b] = \{x : a < x \leq b\}$	half-open
$[a, \infty) = \{x : x \geq a\}$,	$(-\infty, b) = \{x : x < b\}$	unbounded
$(-\infty, \infty) = \mathbb{R}$		the whole line

∞ is *not* a number, only a direction — so it is *always* paired with a parenthesis, never a bracket.

Example 1.5. Solve $|x - 3| < 2$ and write the solution as an interval.

$$|x - 3| < 2 \iff -2 < x - 3 < 2 \iff 1 < x < 5 \iff x \in (1, 5).$$

1.6 Absolute value and distance on the line

The **absolute value** $|x|$ is the distance from x to 0:

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0. \end{cases}$$

The distance between two real numbers is $|a - b|$. Useful facts: $|xy| = |x|\,|y|$, and the **triangle inequality** $|x + y| \leq |x| + |y|$.

Looking ahead. *$|a - b|$ is distance in one dimension. In Unit 2 we generalize it to a* norm *$\|x - y\|$, which measures distance between vectors — and the triangle inequality survives unchanged.*

2. Functions

A **function** $f : A \to B$ assigns to each input $x \in A$ exactly one output $f(x) \in B$. Here A is the **domain**, B the **codomain**, and

$$\text{range}(f) = \{f(x) : x \in A\} \subseteq B$$

is the set of values actually hit (the **image**).

Example 2.1. For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$: the domain is $\mathbb{R}$, the codomain is $\mathbb{R}$, but the range is $[0, \infty)$ because squares are never negative.

Composition feeds one function into another: $(g \circ f)(x) = g\big(f(x)\big)$ — do f first, then g. An **inverse** f^{-1} undoes f: $f^{-1}(f(x)) = x$. A function has an inverse exactly when it is one-to-one (distinct inputs give distinct outputs).

Example 2.2. If $f(x) = 2x+1$ then $f^{-1}(x) = \dfrac{x-1}{2}$, since solving $y = 2x+1$ for x gives $x = \frac{y-1}{2}$. Check: $f^{-1}(f(3)) = f^{-1}(7) = 3$. ✓

Looking ahead. *A* linear transformation *(Unit 8) is a special kind of function between vector spaces, and matrix multiplication is exactly* composition *of such functions. The matrix inverse (Unit 4) is a function inverse. Exponential and logarithm (§5) are inverses of each other.*

3. Summation (Σ) and Product (Π) Notation

3.1 Sigma notation

Σ (capital sigma) is shorthand for a sum. The **index** i runs from the lower bound to the upper bound, and we add up the terms:

$$\sum_{i=1}^{n} a_i \;=\; a_1 + a_2 + \cdots + a_n.$$

The index name is a dummy: $\sum_{i=1}^{n} a_i = \sum_{k=1}^{n} a_k$.

Example 3.1. $\displaystyle\sum_{i=1}^{4} i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 30.$ $\qquad \displaystyle\sum_{k=0}^{3} 2^k = 1 + 2 + 4 + 8 = 15.$

Algebra of sums (use these constantly)

$$\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i \quad \text{(split a sum)}, \qquad \sum_{i=1}^{n} c\,a_i = c\sum_{i=1}^{n} a_i \quad \text{(pull out constants)},$$

$$\sum_{i=1}^{n} c = c\,n \quad \text{(a constant term added } n \text{ times)}, \qquad \textit{Index shift: } \sum_{i=1}^{n} a_i = \sum_{j=0}^{n-1} a_{j+1}.$$

The first two properties together are exactly *linearity*, the single most important structural fact in the whole course.

Example 3.2 (Telescoping)**.** A sum where neighbors cancel:

$$\sum_{i=1}^{n}\bigl(a_{i+1} - a_i\bigr) = a_{n+1} - a_1.$$

For instance $\sum_{i=1}^{n} \frac{1}{i(i+1)} = \sum_{i=1}^{n}\Bigl(\frac{1}{i} - \frac{1}{i+1}\Bigr) = 1 - \frac{1}{n+1}$.

Looking ahead. *The dot product $\langle x, y\rangle = \sum_i x_i y_i$ (Unit 2), the matrix product, and every "cost of an algorithm" count (Unit 6) are written with Σ. Linearity of the sum* is *the reason these objects are called "linear."*

3.2 Pi notation and factorials

Π (capital pi) is the same idea for *products*:

$$\prod_{i=1}^{n} a_i = a_1 \cdot a_2 \cdots a_n.$$

The **factorial** is the product of the first n positive integers:

$$n! = \prod_{i=1}^{n} i = 1 \cdot 2 \cdot 3 \cdots n, \qquad 0! = 1 \text{ (by convention)}.$$

Example 3.3. $\displaystyle\prod_{i=1}^{4} i = 4! = 24. \qquad \prod_{k=1}^{3}(2k) = 2 \cdot 4 \cdot 6 = 48.$

Looking ahead. *The determinant of a matrix is a signed sum of* products *of entries (Unit 4), and equals the* product *of the eigenvalues* $\det A = \prod_i \lambda_i$ *(Unit 11). Factorials reappear in the binomial theorem (§6) and Taylor's theorem (Unit 5).*

4. Common Summation Formulas

These closed forms turn a long sum into a single expression. Memorize the first four; understand where the geometric series comes from.

Closed forms

$$\sum_{i=1}^{n} 1 = n, \qquad \sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Geometric: $\displaystyle\sum_{i=0}^{n} r^i = \frac{1-r^{n+1}}{1-r} \ (r \neq 1), \qquad \sum_{i=0}^{\infty} r^i = \frac{1}{1-r} \ (|r| < 1).$

Arithmetic: $\displaystyle\sum_{i=0}^{n-1}(a + id) = na + d\,\frac{n(n-1)}{2}$ (first term a, common difference d).

Example 4.1 (The Gauss pairing trick)**.** Why is $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$? Write the sum forwards and backwards and add columns:

$$\begin{array}{ccccccccc} S & = & 1 & + & 2 & +\cdots+ & n \\ S & = & n & + & (n-1) & +\cdots+ & 1 \\ \hline 2S & = & (n+1) & + & (n+1) & +\cdots+ & (n+1) \end{array}$$

There are n copies of $(n+1)$, so $2S = n(n+1)$ and $S = \frac{n(n+1)}{2}$. (Legend says Gauss found this as a schoolboy.)

Example 4.2 (Using the formulas)**.**

$$\sum_{i=1}^{10} i = \frac{10 \cdot 11}{2} = 55, \qquad \sum_{i=1}^{5} i^2 = \frac{5 \cdot 6 \cdot 11}{6} = 55, \qquad \sum_{i=0}^{4}\left(\tfrac{1}{2}\right)^i = \frac{1-(1/2)^5}{1-1/2} = \frac{31}{16}.$$

Example 4.3 (Infinite geometric series)**.** $\sum_{i=0}^{\infty}\left(\frac{1}{3}\right)^i = \frac{1}{1-\frac{1}{3}} = \frac{3}{2}$. The series converges only because $|r| = \frac{1}{3} < 1$; for $|r| \ge 1$ it diverges.

Looking ahead. *Summation formulas let us count operations in closed form: a double loop over an $n \times n$ matrix costs $\sum_{i=1}^{n} n = n^2$ steps — this is how we get the "$O(n^2)$" and "$O(n^3)$" costs in Unit 6. The geometric series also governs how fast iterative methods converge (Units 6, 11, 12).*

5. Exponents and Logarithms

5.1 Exponent laws

For a base $a > 0$ and real exponents m, n:

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}, \quad (ab)^n = a^n b^n, \quad a^0 = 1, \quad a^{-n} = \frac{1}{a^n}, \quad a^{1/n} = \sqrt[n]{a}.$$

Example 5.1. $2^3 \cdot 2^4 = 2^7 = 128, \quad \frac{x^5}{x^2} = x^3, \quad (9)^{1/2} = 3, \quad 8^{-2/3} = \frac{1}{8^{2/3}} = \frac{1}{(\sqrt[3]{8})^2} = \frac{1}{4}.$

5.2 Logarithms: the inverse of exponentiation

The logarithm answers "*to what power must I raise the base?*"

$$\log_a x = y \iff a^y = x \qquad (a > 0,\ a \neq 1,\ x > 0).$$

Two special bases: the **natural log** $\ln x = \log_e x$ (base $e \approx 2.718$) and, in computer science, $\log_2$.

Logarithm laws (each mirrors an exponent law)

$\log_a(xy) = \log_a x + \log_a y, \qquad \log_a \frac{x}{y} = \log_a x - \log_a y, \qquad \log_a(x^r) = r \log_a x,$

$\log_a 1 = 0, \qquad \log_a a = 1, \qquad$ *Change of base:* $\log_a x = \frac{\ln x}{\ln a}, \qquad$ *Inverses:* $a^{\log_a x} = x$ and $\log_a(a^x) = x$.

Example 5.2. $\log_2 8 = 3$ since $2^3 = 8$. $\ln(e^5) = 5$. $\log_{10}(1000) = 3$. Using the laws, $\log_2(3 \cdot 8) = \log_2 3 + \log_2 8 = \log_2 3 + 3$.

Example 5.3 (Solving with logs)**.** Solve $5 \cdot 2^x = 160$. Divide: $2^x = 32$, so $x = \log_2 32 = 5$. Solve $e^{3x} = 10$. Take ln: $3x = \ln 10$, so $x = \frac{1}{3} \ln 10 \approx 0.767$.

5.3 What you should recall from MATH 141

You will re-derive these in Unit 5, but keep them handy:

$$\frac{d}{dx} e^x = e^x, \qquad \frac{d}{dx} \ln x = \frac{1}{x} \ (x > 0), \qquad \frac{d}{dx} a^x = a^x \ln a.$$

The number e is special precisely because e^x is its own derivative.

Looking ahead. *Logs turn products into sums — which is why they tame the products of many small probabilities in machine learning (the "log-likelihood" and "cross-entropy loss" of Unit 12). Exponentials define the matrix exponential e^A (Units 4, 11). And $\log_2$ measures how many times you can halve a problem — the source of "$O(\log n)$" running times (Unit 6).*

6. Polynomials

A **polynomial** of degree n is a function

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0,$$

with the a_i constants (*coefficients*) and a_n the *leading* coefficient. A **root** (or *zero*) is a value r with $p(r) = 0$.

Facts about roots

Factor theorem: r is a root $\iff$ $(x - r)$ divides $p(x)$.

Quadratic formula: the roots of $ax^2 + bx + c = 0$ are $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$; the *discriminant* $b^2 - 4ac$ is > 0 (two real roots), $= 0$ (one repeated root), or < 0 (two complex roots).

Counting roots: a degree-n polynomial has *exactly* n roots in $\mathbb{C}$, counted with multiplicity (Fundamental Theorem of Algebra).

Example 6.1 (Factoring)**.** $x^2 - 5x + 6 = (x - 2)(x - 3)$, so the roots are $x = 2$ and $x = 3$. Check with the factor theorem: $p(2) = 4 - 10 + 6 = 0$. ✓

Example 6.2 (Quadratic formula & complex roots)**.** Solve $x^2 + x + 1 = 0$. The discriminant is $1 - 4 = -3 < 0$, so $x = \dfrac{-1 \pm \sqrt{-3}}{2} = -\dfrac{1}{2} \pm \dfrac{\sqrt{3}}{2}\, i$ — two complex roots. (This is why we need $\mathbb{C}$; see Unit 3.)

6.1 The binomial theorem

Expanding $(x + y)^n$ uses **binomial coefficients** $\dbinom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$:

$$(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

Example 6.3. $(x+y)^3 = \binom{3}{0}x^3 + \binom{3}{1}x^2y + \binom{3}{2}xy^2 + \binom{3}{3}y^3 = x^3 + 3x^2y + 3xy^2 + y^3$.

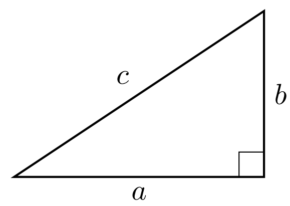
Looking ahead. *The* characteristic polynomial $\det(A - \lambda I)$ *is a degree-n polynomial whose roots are the eigenvalues of a matrix (Unit 11). The Fundamental Theorem of Algebra guarantees n of them (possibly complex, possibly repeated) — which is exactly why eigenvalues can be complex even for a real matrix.*

7. Geometry: Distance and the Circle

7.1 The Pythagorean theorem

In a right triangle with legs a, b and hypotenuse c,

$$a^2 + b^2 = c^2.$$



Example 7.1. Legs 3 and 4: $c = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

7.2 The distance formula

Applying Pythagoras to the horizontal and vertical gaps between two points gives the distance in the plane, and the same idea extends to space:

$$d\big((x_1, y_1), (x_2, y_2)\big) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2},$$
$$d\big((x_1, y_1, z_1), (x_2, y_2, z_2)\big) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Example 7.2. The distance from $(1, 2)$ to $(4, 6)$ is $\sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9 + 16} = 5$.

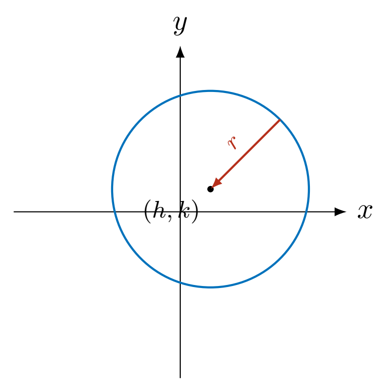
Looking ahead. *This is the* formula the course generalizes most. *The Euclidean norm* $\|x\|_2 = \sqrt{\sum_i x_i^2}$ *(Unit 2) is just the distance formula in* n *dimensions, and least squares (Unit 10) minimizes exactly this distance between a model and the data.*

7.3 The equation of a circle

A circle is the set of points at a fixed distance r (the *radius*) from a center (h, k). Setting "distance to center $= r$" and squaring:

$$(x - h)^2 + (y - k)^2 = r^2.$$

Centered at the origin this is $x^2 + y^2 = r^2$. The 3-D analog is the **sphere** $(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$.



Example 7.3. $(x - 1)^2 + (y + 2)^2 = 9$ is the circle with center $(1, -2)$ and radius 3. Conversely, complete the square on $x^2 + y^2 - 6x + 4y + 9 = 0$:

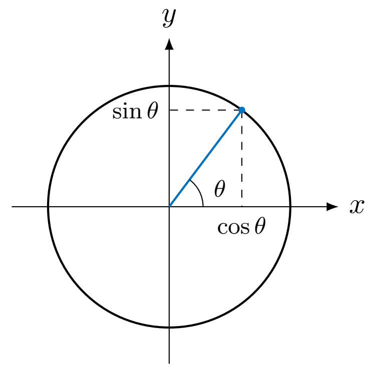
$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = -9 + 9 + 4 \;\Rightarrow\; (x - 3)^2 + (y + 2)^2 = 4,$$

a circle of center $(3, -2)$, radius 2.

Looking ahead. *"All points at distance* $\leq r$*" is a disk — and in* n *dimensions, the set* $\{x : \|x\| \leq r\}$ *is the* unit ball *of a norm (Unit 2). Its shape (round for* ℓ_2*, diamond for* ℓ_1*, square for* ℓ_∞*) explains why different norms behave so differently in regularization and optimization (Unit 12).*

7.4 A little trigonometry and the unit circle

On the **unit circle** $x^2 + y^2 = 1$, the point at angle θ (measured counterclockwise from the positive x-axis) is $(\cos\theta, \sin\theta)$.



Because the point lies on the circle, we get the **Pythagorean identity**

$$\cos^2\theta + \sin^2\theta = 1,$$

and $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$. Angles are measured in **radians**, where $180° = \pi$; a few values worth knowing:

$$\cos 0 = 1,\ \sin 0 = 0; \quad \cos\tfrac{\pi}{2} = 0,\ \sin\tfrac{\pi}{2} = 1; \quad \cos\tfrac{\pi}{4} = \sin\tfrac{\pi}{4} = \tfrac{\sqrt{2}}{2}.$$

Looking ahead. *The angle between two vectors (Unit 2) is measured with* $\cos\theta$*; cosine* similarity *in machine learning is exactly this* $\cos\theta$*. Rotations in the plane, complex multiplication (Unit 3), and quaternions all run on* $\sin$ *and* $\cos$*.*

7.5 Lines in the plane

A non-vertical line has **slope** $m = \dfrac{\Delta y}{\Delta x} = \dfrac{y_2 - y_1}{x_2 - x_1}$ and can be written

$$y = mx + b \quad \text{(slope–intercept)}, \qquad y - y_1 = m(x - x_1) \quad \text{(point–slope)}.$$

Example 7.4. The line through $(1,2)$ and $(3,8)$ has slope $m = \frac{8-2}{3-1} = 3$, so $y - 2 = 3(x-1)$, i.e. $y = 3x - 1$.

Looking ahead. *A line is the graph of a* linear *equation in two variables. A* system *of such equations is the starting point of the entire course (Unit 7), and fitting the "best" line through scattered data is linear regression (Unit 10).*

8. Greek Alphabet and Symbol Glossary

Linear algebra borrows heavily from Greek. You do not need to memorize this — just know where to look.

Symbol	Name	Symbol	Name
α	alpha	λ	lambda (*eigenvalues*)
β	beta	μ	mu (mean)
γ	gamma	$\sigma,\ \Sigma$	sigma (*singular values / sum*)
$\delta,\ \Delta$	delta (small change)	$\pi,\ \Pi$	pi (*product*)
ε	epsilon (tiny quantity)	ϕ, φ	phi
θ	theta (angle)	ψ	psi
κ	kappa (*condition number*)	$\omega,\ \Omega$	omega

Symbol	Meaning	Symbol	Meaning
$\in,\ \notin$	is / is not an element of	$\forall$	for all
$\subseteq,\ \subsetneq$	subset / proper subset	$\exists$	there exists
$\cup,\ \cap$	union / intersection	$\Rightarrow$	implies
$\varnothing$	empty set	$\iff$	if and only if
$\mathbb{R},\ \mathbb{C},\ \mathbb{Z}$	reals, complex, integers	$\approx$	approximately equals
$\lvert x\rvert,\ \lVert x\rVert$	absolute value / norm	$\therefore$	therefore

9. Self-Check (answers below)

Work these with a pencil; if any feel shaky, reread the matching section.

1. Write $\{x \in \mathbb{Z} : -2 \le x < 3\}$ in roster form.
2. With $A = \{1, 2, 3, 4\},\ B = \{2, 4, 6\}$, find $A \cup B$, $A \cap B$, $A \setminus B$.
3. Solve $|2x - 1| \le 5$ and write the answer as an interval.
4. Evaluate $\displaystyle\sum_{i=1}^{6} i$ and $\displaystyle\sum_{i=1}^{4} i^2$.
5. Evaluate $\displaystyle\sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^k$.
6. Simplify $\log_2 40 - \log_2 5$.
7. Solve $3^x = 81$ and $e^{2x} = 7$ (leave the second exact).
8. Find the roots of $x^2 - 4x + 13 = 0$.
9. Find the distance from $(-1, 3)$ to $(2, -1)$.
10. Give the center and radius of $x^2 + y^2 + 4x - 6y + 9 = 0$.

Answers

1. $\{-2, -1, 0, 1, 2\}$.
2. $A \cup B = \{1, 2, 3, 4, 6\},\ \ A \cap B = \{2, 4\},\ \ A \setminus B = \{1, 3\}$.

3. $-5 \le 2x - 1 \le 5 \Rightarrow -2 \le x \le 3$, i.e. $[-2, 3]$.
4. $\sum_{i=1}^{6} i = \frac{6 \cdot 7}{2} = 21$; $\sum_{i=1}^{4} i^2 = \frac{4 \cdot 5 \cdot 9}{6} = 30$.
5. $\frac{1}{1 - 1/4} = \frac{4}{3}$.
6. $\log_2 \frac{40}{5} = \log_2 8 = 3$.
7. $x = 4$ (since $3^4 = 81$); $x = \frac{1}{2} \ln 7$.
8. Discriminant $16 - 52 = -36$, so $x = \frac{4 \pm 6i}{2} = 2 \pm 3i$.
9. $\sqrt{(2 - (-1))^2 + (-1 - 3)^2} = \sqrt{9 + 16} = 5$.
10. Complete the square: $(x + 2)^2 + (y - 3)^2 = 4$; center $(-2, 3)$, radius 2.

That is the whole toolkit. If the self-check went well, you are ready for Unit 1. Keep this sheet nearby — every symbol here reappears in the first three weeks.